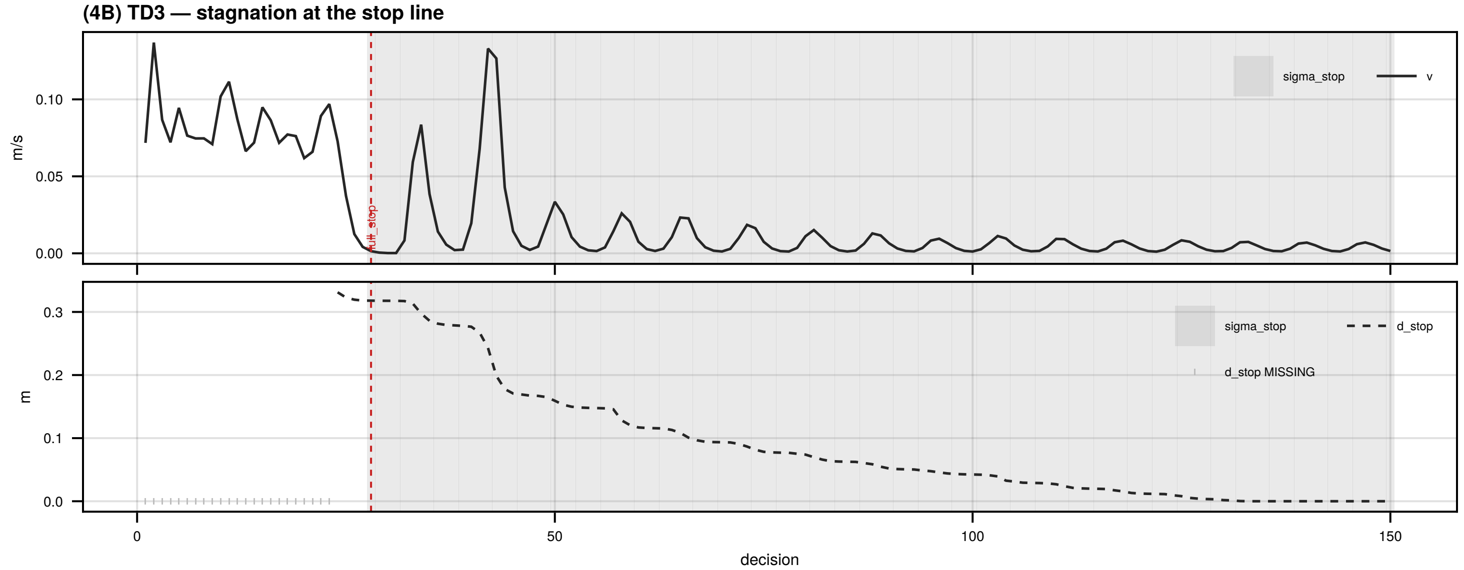
Figure 4. Episode behaviour and failure mechanisms

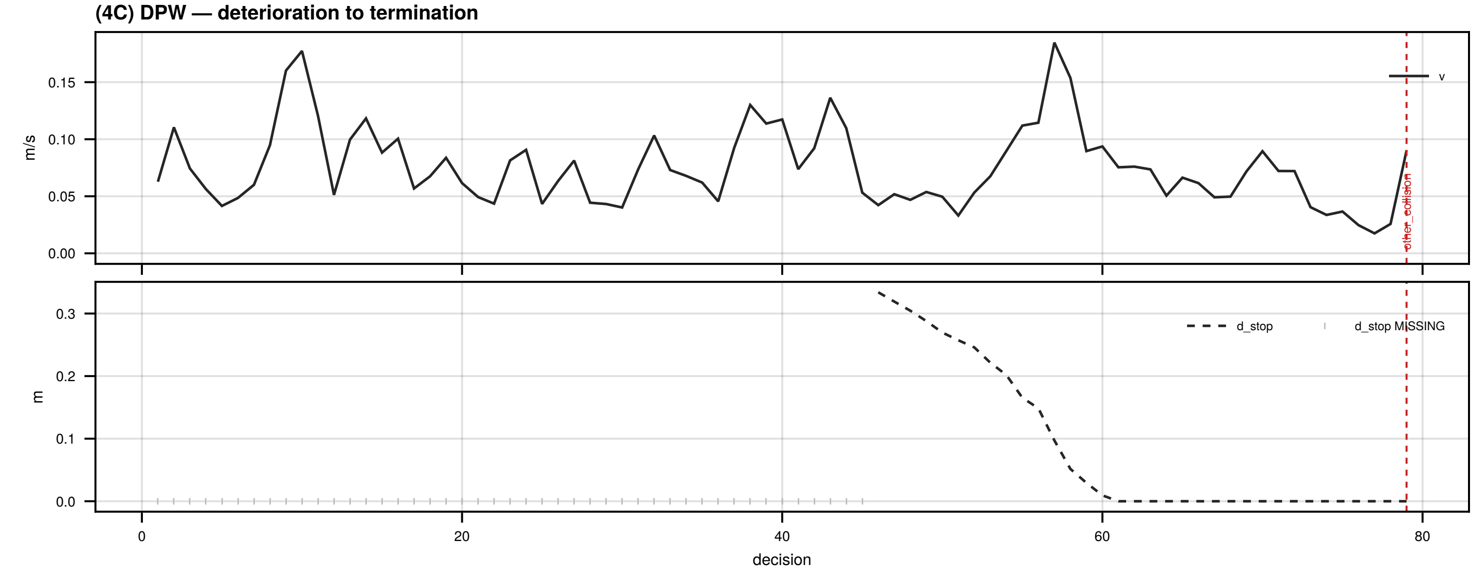
(4A) Six-solver outcome summary

solver	eps	mean ret	med ret	len	env end	horizon	offroad	collide	stop enc	compliance
dpw@1k	20	-214.46	-211.59	85.6	19	1	13	6	7	0.0%
mcts@1k	20	6.78	7.75	150.0	0	20	0	0	4	100.0%
q_learning	20	11.63	12.86	145.6	1	19	1	0	8	100.0%
sac	20	-1.32	4.75	150.0	0	20	0	0	20	90.0%
sarsa	20	11.92	12.86	144.8	1	19	1	0	8	100.0%
td3	20	-213.78	-216.08	150.0	0	20	0	0	0	N/A

(4B) TD3 — stagnation at the stop line



(4C) DPW — deterioration to termination



(4D) Generative calls by episode progress

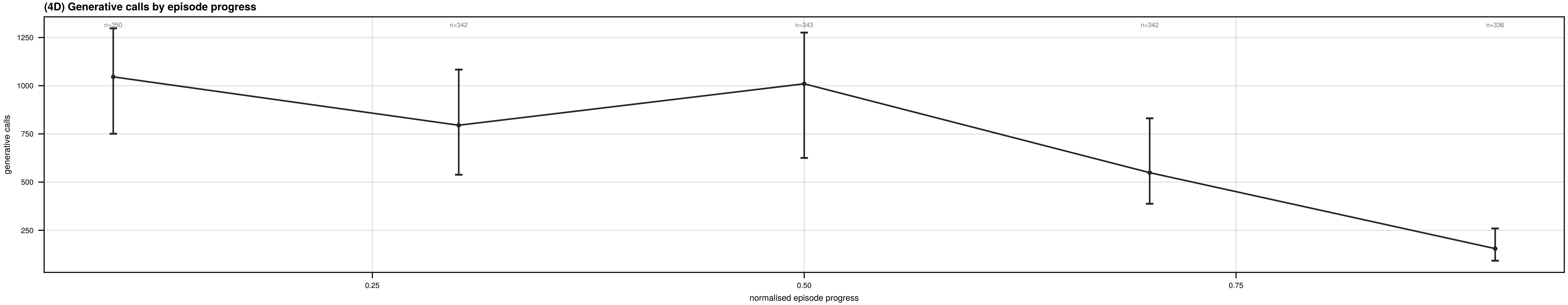


Figure 4. Episode behaviour and failure mechanisms. (4A) The frozen six-solver comparison over 20 paired seeds; task performance and computational cost are reported on separate axes and never combined into one score. (4B) TD3, seed 1001: it reaches the stop sign and performs a full stop at decision 28, then remains in the stop zone for 123 of 150 decisions at a mean speed of 0.005 m/s and never proceeds past it — `passed_stops = 0` across all 20 seeds — so the episode ends only when the evaluation horizon is reached, with the return dominated by the stagnation penalty. (4C) DPW, seed 1002: navigation degrades until the environment terminates the episode at decision 79 (other_collision). (4D) Realised generative calls per decision by normalised episode progress for DPW: 1019 → 835 → 940 → 602 → 195. Compute is endogenous — it falls as the trajectory deteriorates and the planner has less tree to expand. Episodes were selected by stated rule (td3 first_stagnation seed 1001; dpw@1k median_length_terminating seed 1002; mcts@1k median_return seed 1018), not by inspection.

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